Name:_____

Unfortunately, it's impossible to solve most differential equations in the sense of obtaining an explicit formula for the solution. Here we will see that, despite the absence of an explicit solution, we can still learn a lot about the solution through a graphical approach.

Suppose we are asked to sketch the graph of a solution of the initial-value problem

$$y' = x + y, \quad y(0) = 1. \tag{1}$$

In this case we don't know a formula for the solution $y(x)$, so we can't use it to sketch the graph. Instead, we think about what the differential equation in (1) means. Equation (1) says that the slope of the tangent line to the graph at the point (x, y) is equal to the sum of the x and y coordinates of the point. For example, the initial condition says the curve passes through $(0, 1)$, and the differential equation says the slope of the tangent line to the curve at $(0, 1)$ is equal to $0 + 1 = 1$. We draw a small segment of the tangent line at this point, which we can think of, to good approximation, as a small portion of the curve itself.

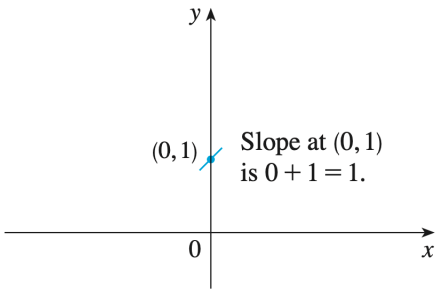
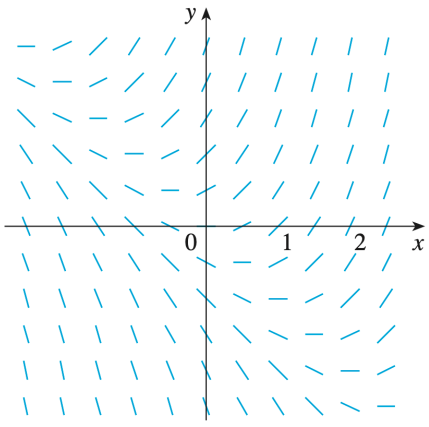
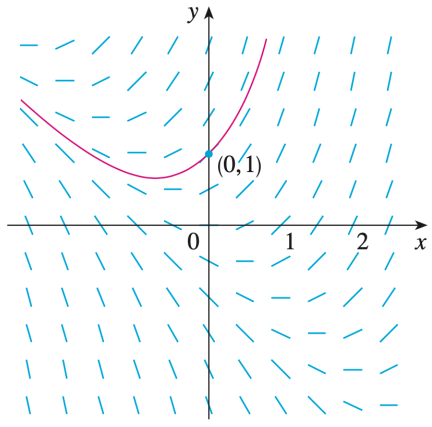


Figure 1: A small portion of the solution curve to $y' = x + y,\ y(0) = 1$.

Now calculate the slope at several other points (x, y) in the plane and plot short line segments with this slope for each point. This plot is called the *direction field* (or *slope field*) of the differential equation, as shown to the left below. The direction field allows us to visualize the general shape of the solution curves by indicating the direction which the curves proceed at each point. We then sketch the solution curve through the point $(0, 1)$ by following the direction field, as shown to the right below. Notice that we have drawn the curve so that it is parallel to nearby line segments.



(a) The direction field for $y' = x + y$.



(b) The solution curve for $y' = x + y,\ y(0) = 1$.

Name:_____

Match each direction field with the correct differential equation.

(1) $y' = y + 2$

(2) $y' = x^2 + 1$

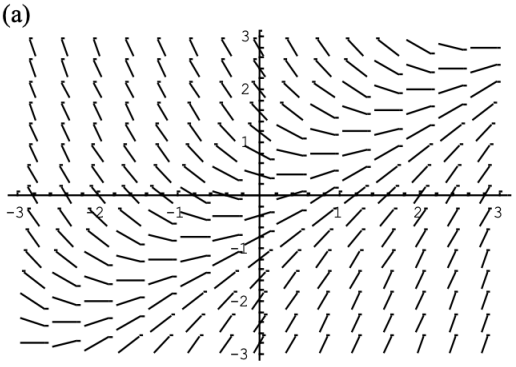
(3) $y' = x(2 - y)$

(4) $y' = \sin(x)$

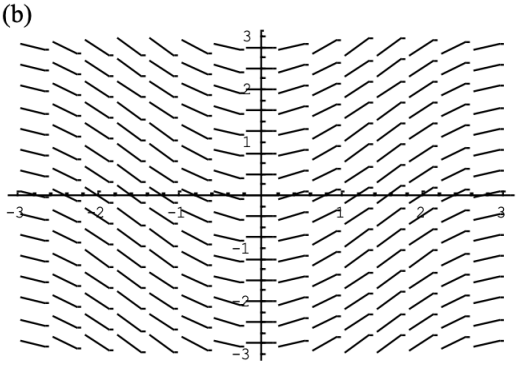
(5) $y' = x - y$

(6) $y' = y(2 - y)$

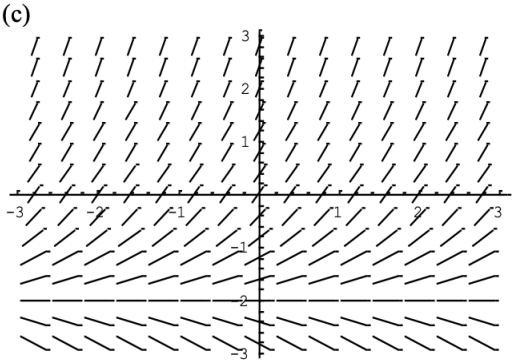
(a)



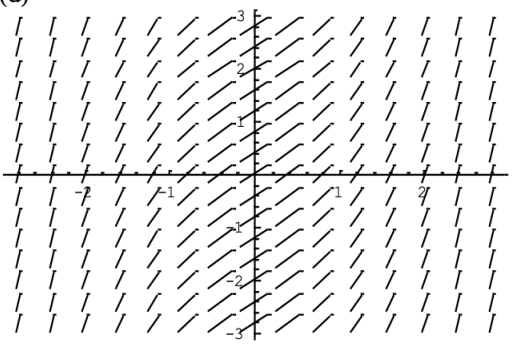
(b)



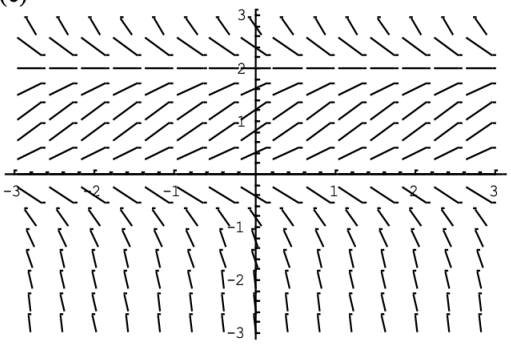
(c)



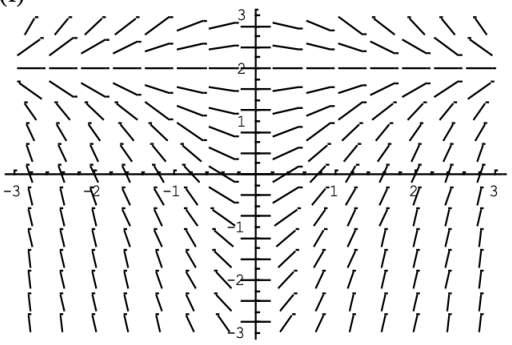
(d)



(e)



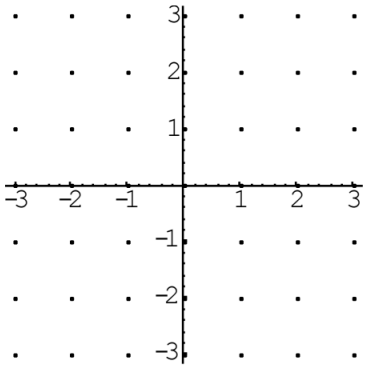
(f)



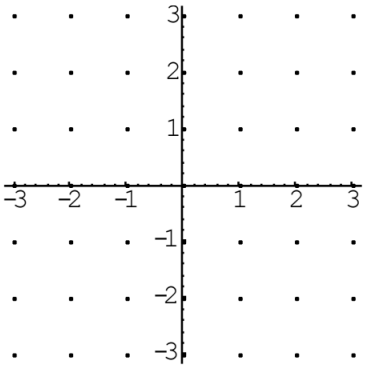
Name:_____

Pick your favorite differential equation below and sketch the slope field at the indicated points. If you are feeling bored or need more practice, you can do more than one.

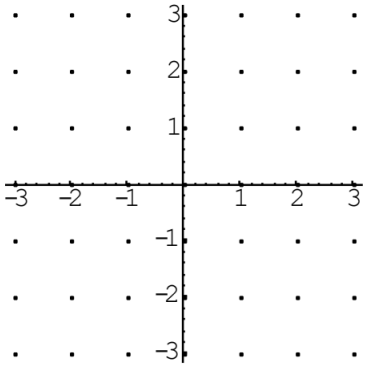
1. $y' = \frac{x}{2}$



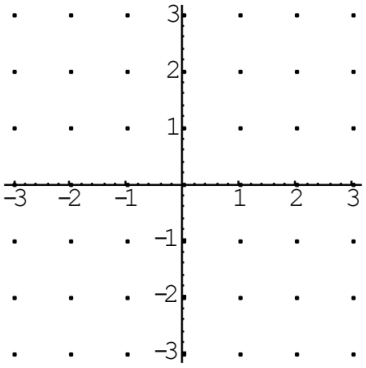
2. $y' = \frac{y}{2}$



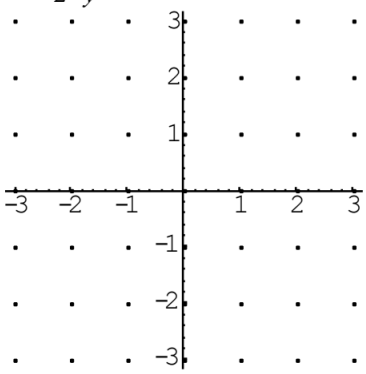
3. $y' = x + \frac{y}{2}$



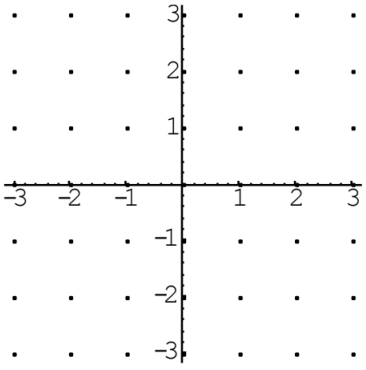
4. $y' = \frac{1}{2}xy$



5. $y' = \frac{1}{2} \cdot \frac{x}{y}$



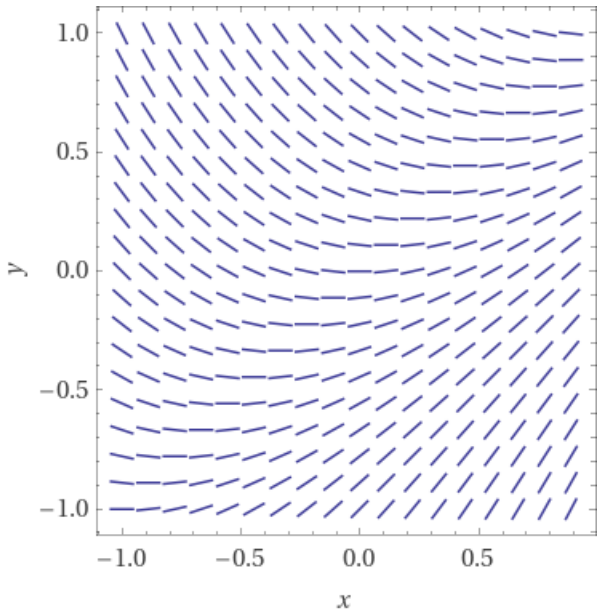
6. $y' = \frac{y}{2}(1 + x)$



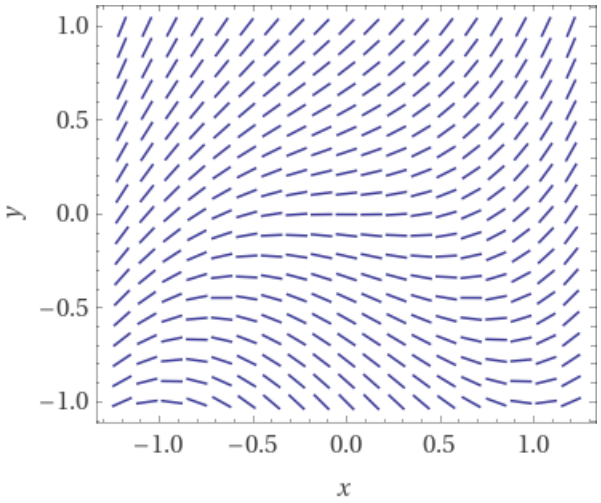
 Name:_____

For each initial-value problem, use the direction field to sketch the solution curve. Be sure to indicate which curve corresponds to which initial condition.

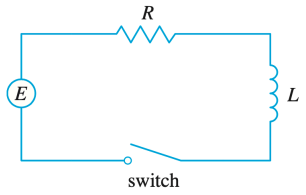
1. $y' = x - y$
   a. $y(0) = -3$ b. $y(-1) = -2$ c. $y(-3) = 0$



2. $y' = x^2 + y$
   a. $y(0) = 1$ b. $y(0) = -2$ c. $y(2) = -2$



The next example shows how direction fields can give insight into physical processes. The electric circuit shown below contains an electromotive force (usually a battery or generator) that produces a voltage of $E(t)$ volts (V) which drives a current of $I(t)$ amperes (A) at time t. The circuit also contains a resistor with a resistance of R ohms (Ω) and an inductor with an inductance if L henries (H).



Ohm's Law gives the drop in voltage due to the resistor as RI. The voltage drop due to the inductor is $L\frac{dI}{dt}$. One of Kirchhoff's laws says that the sum of the voltage drops is equal to the supplied voltage $E(t)$. Thus we have

$$L\frac{dI}{dt} + RI = E \tag{2}$$

which is a first-order differential equation that models the current I at time t.

Suppose that in the circuit above the resistance is $R = 12\ \Omega$, the inductance is $L = 4$H, and the battery provides a constant voltage of $E = 60$V.

(a) Show that, with these values, Equation (2) can be written as $\frac{dI}{dt} = 15 - 3I$.

(b) Draw a direction field for the differential equation in part (a).

(c) What can you say about the current at late times (i.e., what is $\lim_{t\to\infty} I(t)$)?

(d) Identify any equilibrium solutions.

(e) If the switch is closed when $t = 0$ so that the current starts with $I(0) = 0$, use the direction field to sketch the solution curve.